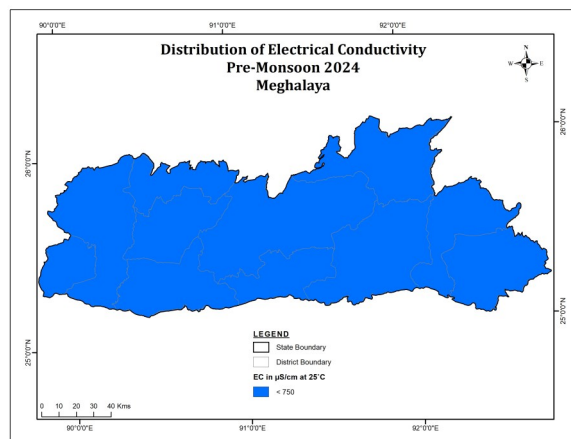


Groundwater Quality Scenario in Meghalaya

Parameters	No of samples	Permissible limit	No. of Samples above permissible limit	% Samples above permissible Limit
EC	39	3000 $\mu\text{S}/\text{cm}$	0	0
Fluoride	39	1.5 mg/L	0	0
Nitrate	39	45 mg/L	0	0
Arsenic	38	10 ppb	0	0
Uranium	38	30 ppb	0	0



Districts with anomalous values at sporadic locations

EC (3000 $\mu\text{S}/\text{cm}$)	Not Any
Fluoride (F > 1.5 mg/L)	Not Any
Nitrate (Nitrate > 45 mg/L)	Not Any
Arsenic (As > 10 ppb)	Not Any
Uranium (U > 30 ppb)	Not Any

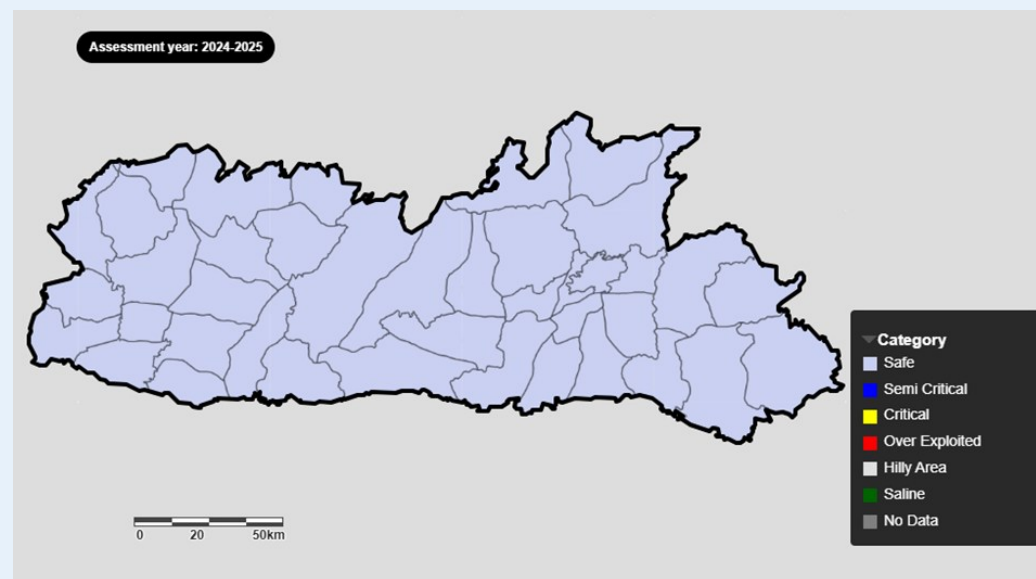
For Further Information, Contact to :
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NH IV Faridabad, Haryana - 121001
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Central Ground Water Board Department of Water Resources, RD & GR Ministry of Jal Shakti, Government of India



Dynamic Ground Water Resources, Ground Water Level & Ground Water Quality of Meghalaya, 2025

November, 2025

Groundwater Resource Scenario in Manipur

- ◆ Ground Water Resources Assessment (GWRA)- jointly carried out by Central Ground Water Board and State Nodal/Ground Water Department periodically as per the Ground Water Resource Estimation Committee (GEC) methodology.
- ◆ Carried out under the guidance of the respective State/UT Level Committees (SLCs) and overall supervision of Central Level Expert Group (CLEG).
- ◆ As part of the assessment, 'Annual Extractable Ground Water Resource' as well as 'Annual Ground Water Extraction' are assessed for each assessment unit (Block).
- ◆ The 'Stage of Ground Water Extraction' is computed as the ratio of 'Annual Ground Water Extraction' with respect to 'Annual Extractable Ground Water Resource' and is usually expressed in percentage. Based on the stage of extraction, the assessment units are categorized as Safe ($\leq 70\%$), Semi-Critical ($>70\%$ and $\leq 90\%$), Critical ($>90\%$ and $\leq 100\%$) and Over-Exploited ($>100\%$).
- ◆ GWRA- 2025, 2024, 2023, 2022 and 2020 has been carried out through a software/web-based application "INDIA-GROUNDWATER RESOURCE ESTIMATION SYSTEM (IN-GRES)" developed by CGWB

Salient Features

1	Average Annual Rainfall	3,074.77 mm
2	Hydrogeology	The state is occupied by hard rocks belonging to the Archean gneissic complex with acidic and basic intrusives and Precambrian Shillong Group of para metamorphites. Ground water occurs under unconfined condition in the weathered residuum and fractured rocks.
3	Recharge Worthy Area of the State	8.14 Thousand Sq. Km
4	Assessment Unit (AU) Type / Number	Block / 40 Numbers
5	Average area of Assessment Unit	203.39 Sq. Km

Findings

	Attribute	GWRA-2017	GWRA-2020	GWRA-2022	GWRA-2023	GWRA-2024	GWRA-2025
1	Total Annual Ground Water Recharge (in bcm)	1.83	2.04	1.72	1.83	1.86	1.84
2	Annual Extractable Ground Water Resources (in bcm)	1.64	1.82	1.51	1.51	1.53	1.54
3	Annual Ground Water Extraction (in bcm)	0.04	0.08	0.05	0.07	0.07	0.08
4	Stage of Ground Water Extraction (in %)	2.28	4.22	3.55	4.58	4.60	5.24

bcm: Billion Cubic Meters

Categorization of Assessment Units based on the 'Stage of Ground Water Extraction'

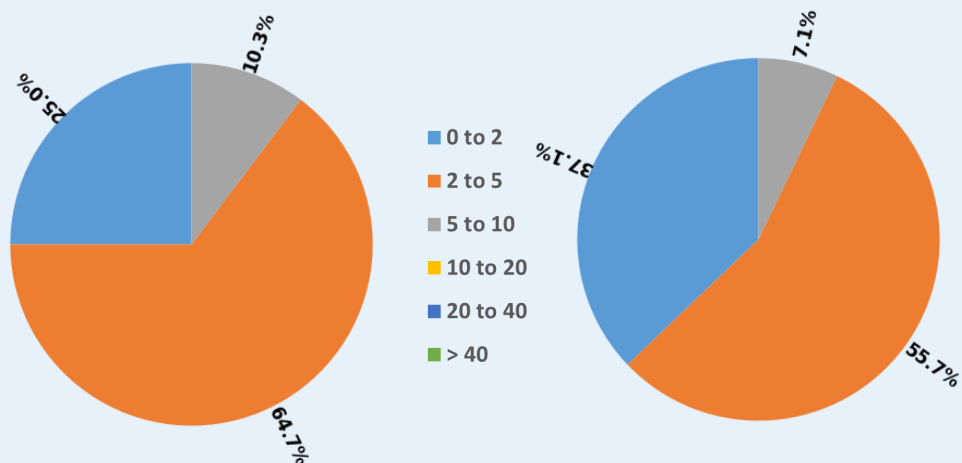
S. No	Category	GWRA-2017		GWRA-2020		GWRA-2022		GWRA-2023		GWRA-2024		GWRA-2025	
		Number of AUs	% of AUs	Number of AUs	% of AUs	Number of AUs	% of AUs	Number of AUs	% of AUs	Number of AUs	% of AUs	Number of AUs	% of AUs
1	Safe	11	100	12	100	12	100	39	100	40	100	40	100
2	Semi-critical												
3	Critical												
4	Over-exploited												
5	Saline												
Total number of AUs		11		12		12		39		40		40	

Recommendations: -

- * The Meghalaya State is essentially occupied by hard rocks belonging to the Archean gneissic complex with acidic and basic intrusive and Precambrian Shillong Group of Para metamorphites. Ground water occurs under unconfined condition in the weathered residuum and fractured rocks and restricted to about 150 m depth. The south-western, southern and south-eastern parts of the state is covered by semi- consolidated formations comprising sandstones, shales, conglomerates, limestones etc. belonging to Cretaceous – Tertiary age. The aquifers are formed by rock strata that are granular/porous, fissured/fractured or cavernous. These aquifers are thick and discontinuous in nature. The unconsolidated sediments comprising sand, gravel, silt, clay, etc. are found to occur as thin veneer along rivulets and as valley-fills.
- * The Ground water resources have been assessed block-wise. The Total Annual Ground Water Recharge of the State has been assessed as 1.84 bcm and Annual Extractable Ground Water Resources as 1.54 bcm. The current Annual Ground Water Extraction is 0.08 bcm and Stage of Ground Water Extraction is 5.24 %.
- * Out of 40 assessment units, all are categorized as 'Safe' assessment units .
- * All the assessment units are in safe category as well as future allocation of ground water is also sufficient, State Government can judiciously develop the ground water resource mainly for agricultural use, however, at no point of time the extraction level should exceed 70%.
- * Development of springs and their catchment in hilly areas for their sustainability.
- * National Aquifer Mapping & Management Programme (NAQUIM) Reports prepared by CGWB (<https://cgwb.gov.in/cgwbpm/>) which are being shared with State/District Authorities and Ground Water Year Book published by CGWB having water level & water quality data may be used in Ground water management (<https://cgwb.gov.in/cgwbpm/>).
- * Regulation & control of Ground water Extraction: Ministry of Jal Shakti has issued the guidelines for control and regulations of ground water extraction vide notification dated 24.09.2020 which has further been amended in March 2023. Concerned departments may ensure implementations of the guidelines.

Groundwater Level Scenario in Meghalaya

1. Depth to Groundwater Level

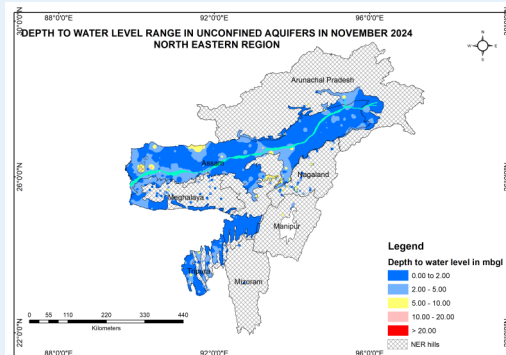
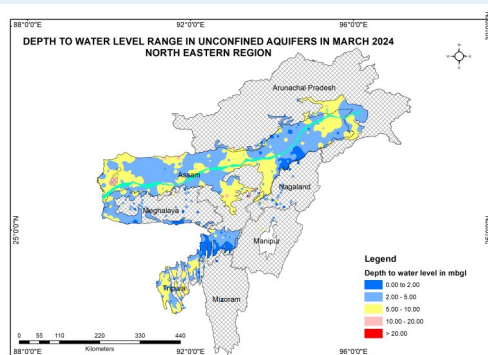


Pre-Monsoon (May 2024)

About 26% of wells recorded groundwater levels in 0–2 m bgl, 64% in 2–5 m bgl, and 10% in 5–10 m bgl, with overall levels ranging from 0.3 m to 8.7 m bgl, indicating predominantly shallow groundwater.

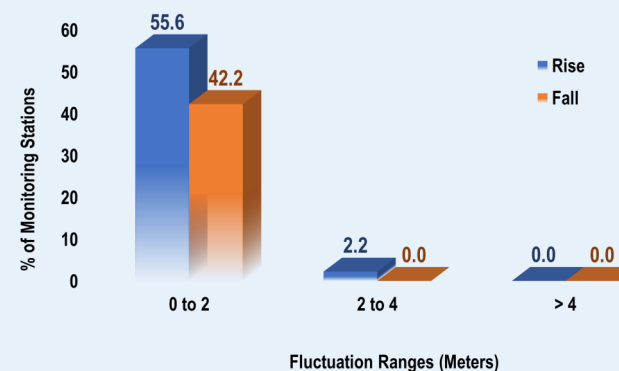
Post-Monsoon (November 2024)

About 41% of wells recorded groundwater levels in 0–2 m bgl, 55% in 2–5 m bgl, and 4% in 5–10 m bgl, with overall levels ranging from 0.2 m to 6.83 m bgl, indicating predominantly shallow groundwater.



2. Decadal Fluctuation of Groundwater Levels

Pre-Monsoon (May 2024 vs Decadal Mean: May 2014–2023)



Post-Monsoon (November 2024 vs Decadal Mean: November 2014–2023)

